

Analysis group meeting

Mathieu Ouillon

5/19/25

IBCuSn Pass0v11 (skimmed)

Number of events: 10,693,979

Number of events with electron: 8658469, percentage: 80.96583133368786

Number of events with electron and pion plus: 2596348, percentage: 24.278596395223893

Number of events with electron and pion minus: 1146175, percentage: 10.717946986804444

Number of events with electron and pion plus and pion minus: 436385, percentage: 4.08066071571676

IBLD2 Pass0v11 (skimmed)

Number of events: 9,259,311

Number of events with electron: 8886695, percentage: 95.97576968739898

Number of events with electron and pion plus: 3311913, percentage: 35.768460525842585

Number of events with electron and pion minus: 1447822, percentage: 15.636390223851429

Number of events with electron and pion plus and pion minus: 624923, percentage: 6.749130685857

IBCxC Pass0v11 (skimmed)

Number of events: 4,618,607

Number of events with electron: 4264301, percentage: 92.32872595568318

Number of events with electron and pion plus: 1490617, percentage: 32.27416838020641

Number of events with electron and pion minus: 635248, percentage: 13.754103780642085

Number of events with electron and pion plus and pion minus: 265646, percentage: 5.751647628819

LD2 Outbending Pass0v11

Number of events: 286,579,714

Number of events with electron: 268,282,597, percentage: 93.61% (Vz cuts)

Number of events with electron and pion plus: 50,006,757, percentage: 17.44%

Number of events with electron and pion minus: 74,707,283, percentage: 26.06%

Number of events with electron and pion plus and pion minus: 16,686,965, percentage: 5.82%

OBCuSn Pass0v11

Number of events: 614,184,230

Number of events with electron: 489079928, percentage : 79.63081826441555

Number of events with electron and pion plus: 67558215, percentage : 10.999666175082352

Number of events with electron and pion minus: 96413429, percentage : 15.6978027586283

Number of events with electron and pion plus and pion minus: 19160077, percentage; 3.11959

CxC Outbending Pass0v11

Number of events: 231,493,276

Number of events with electron: 206224941, percentage: 89.08%

Number of events with electron and pion plus: 33905811, percentage : 14.64%

Number of events with electron and pion minus: 49294898, percentage : 21.29%

Number of events with electron and pion plus and pion minus: 10580974, percentage : 4.57%

Same or more stats for outbending compared to 3 runs
Less stats for inbending

Compared to 3 runs at 100M each -> $300 * 0.7 = 210M$

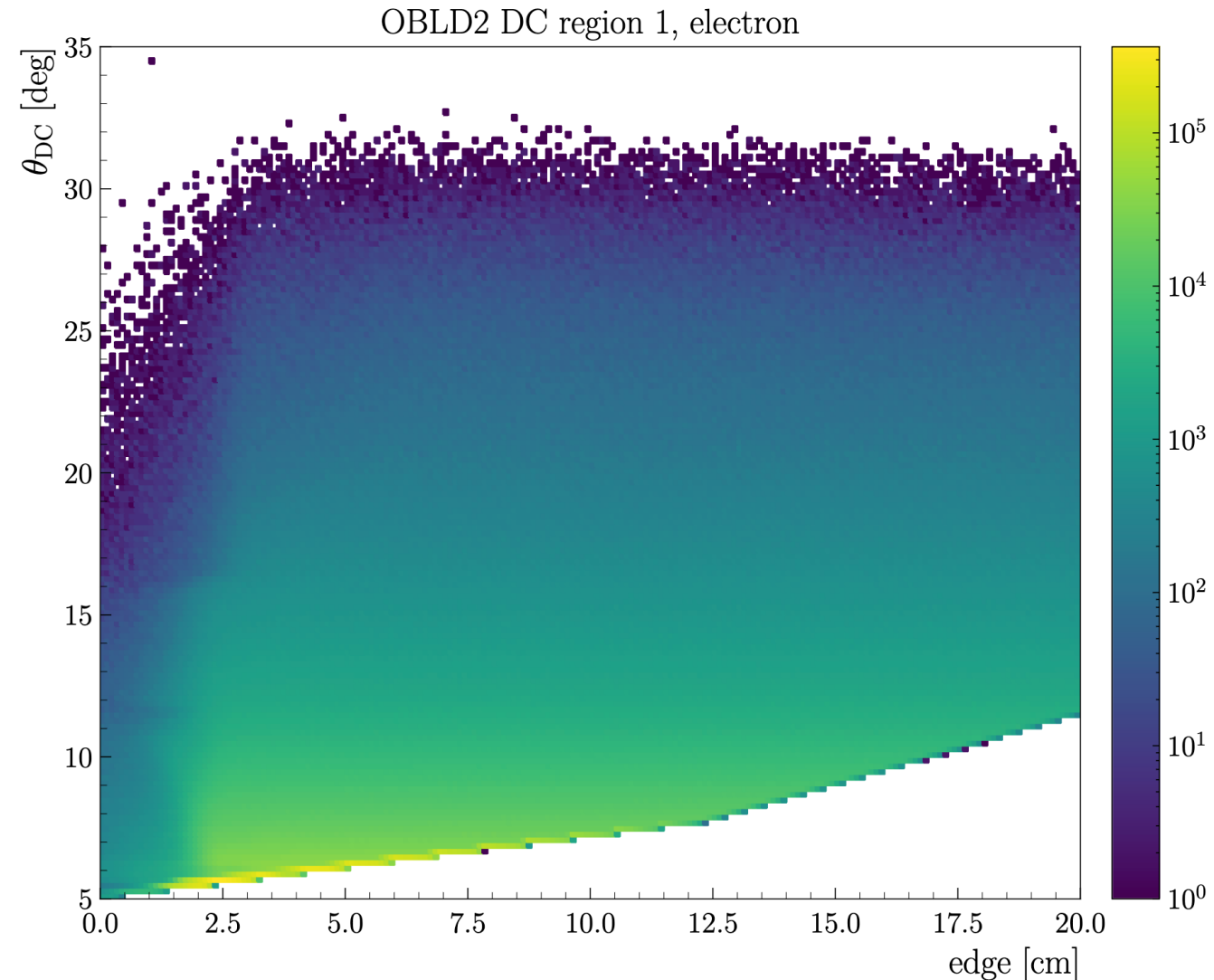
- Size skim forward: 11:X+:X-:Xn 143TB
- Size skim forward: 11:X+:X-:Xn, $Q^2 > 0.7 \text{ GeV}^2$ and $W > 1.8 \text{ GeV}$ 98.9TB
- Removing the RICH bank (RICH::Particle, RICH::Ring, RICH::tdc) and REC::Caloextra and REC::ScintaExtra lead to a 29% reduction
 - 11:X+:X-:Xn 102TB
 - 11:X+:X-:Xn && $Q^2 > 0.7 \text{ GeV}^2$ && $W > 1.8 \text{ GeV}$ 70.2TB

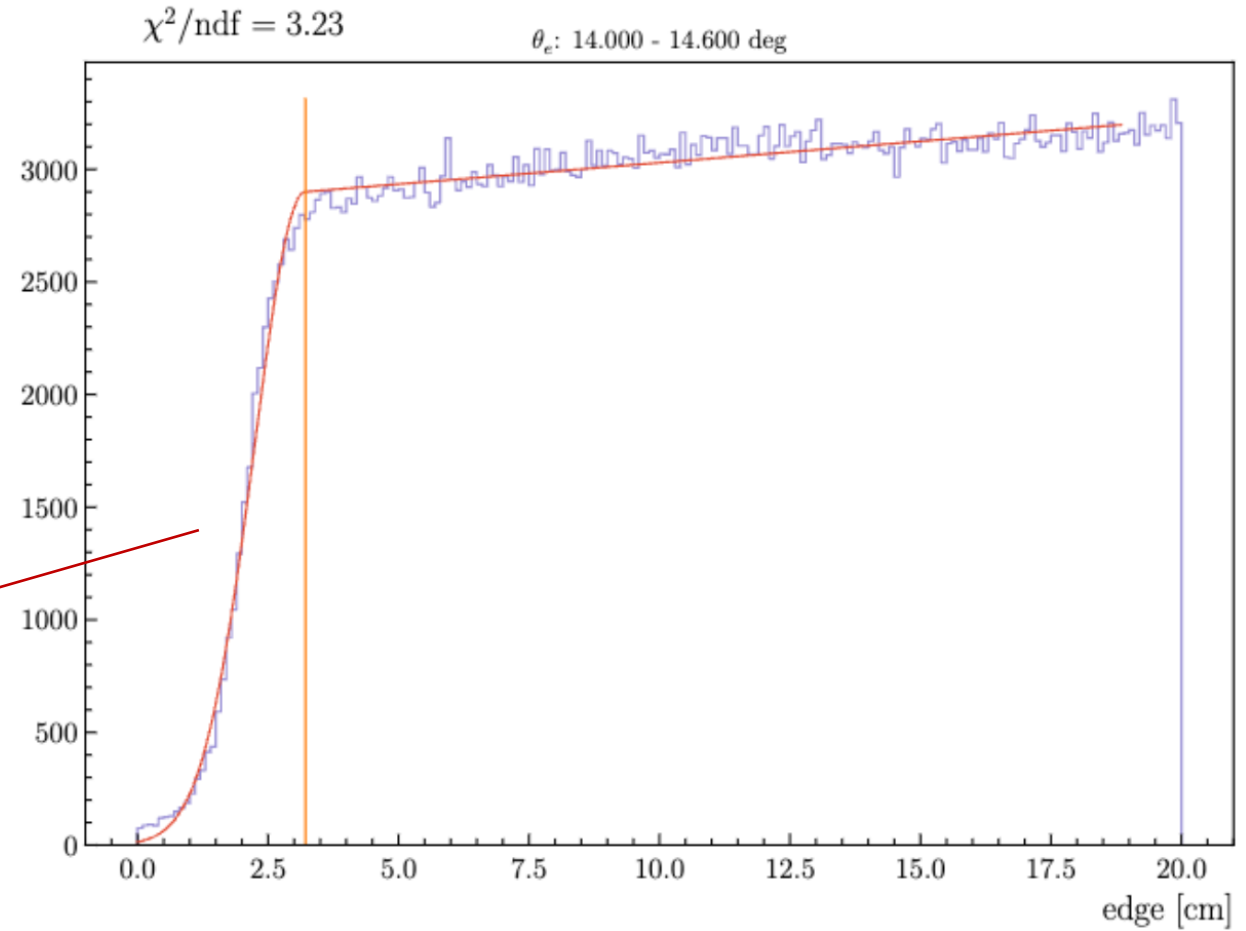
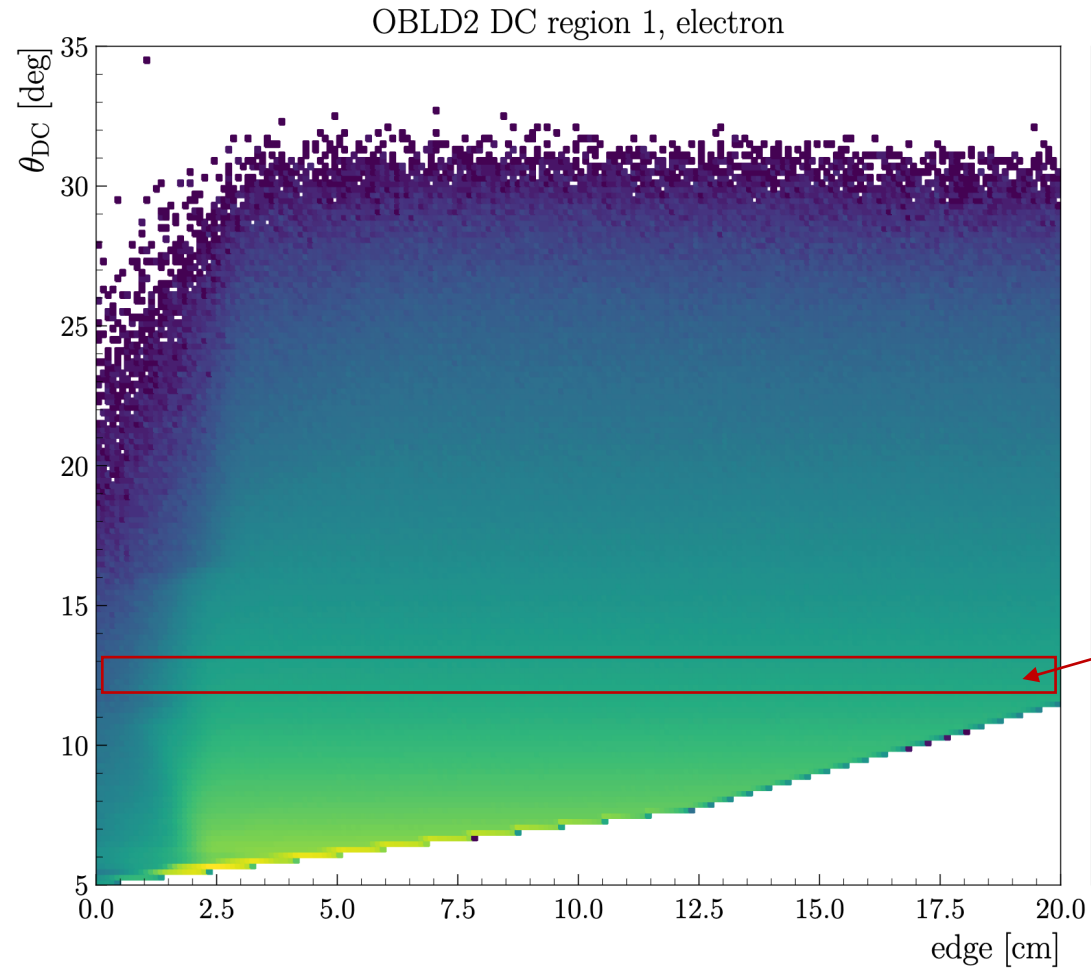
RG-D: DC Fiducial cuts

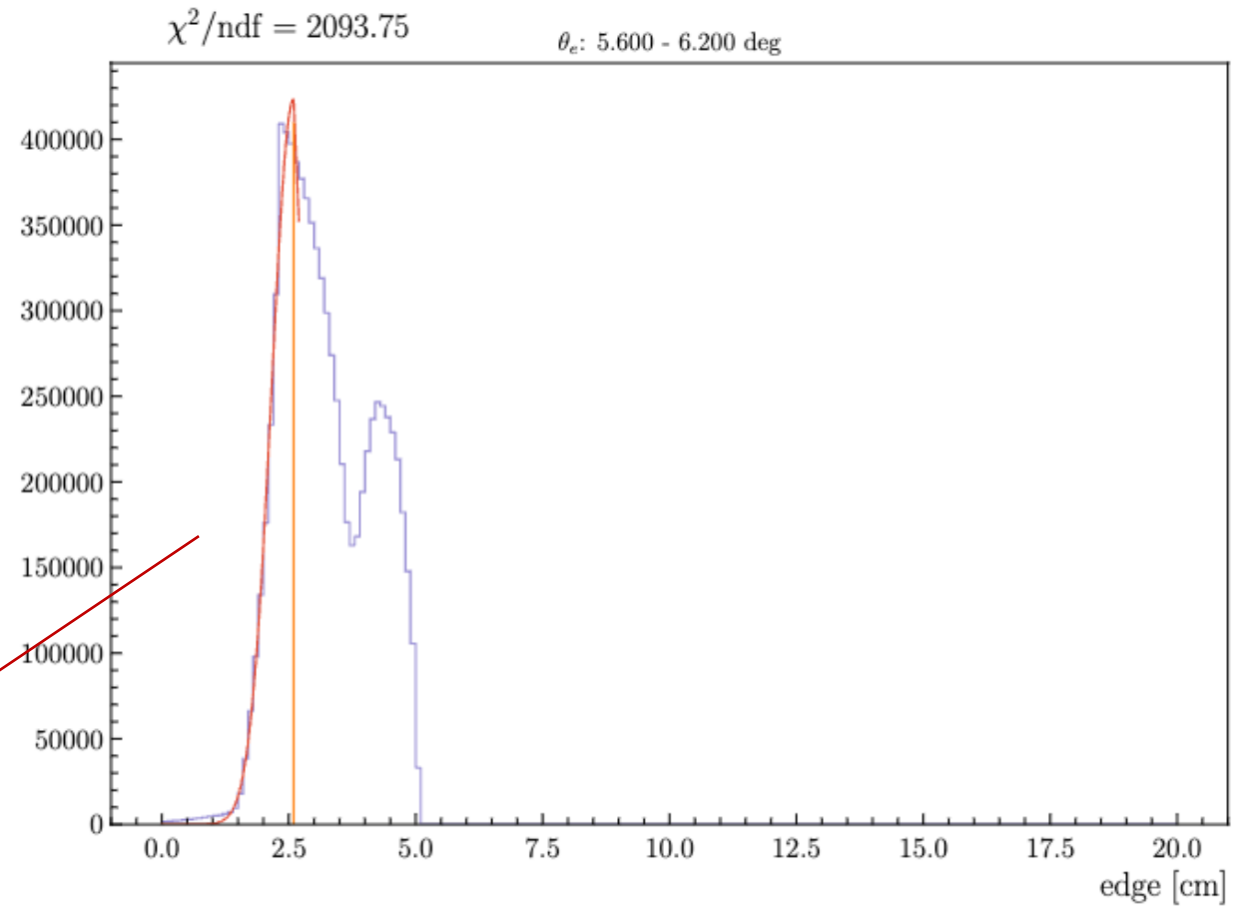
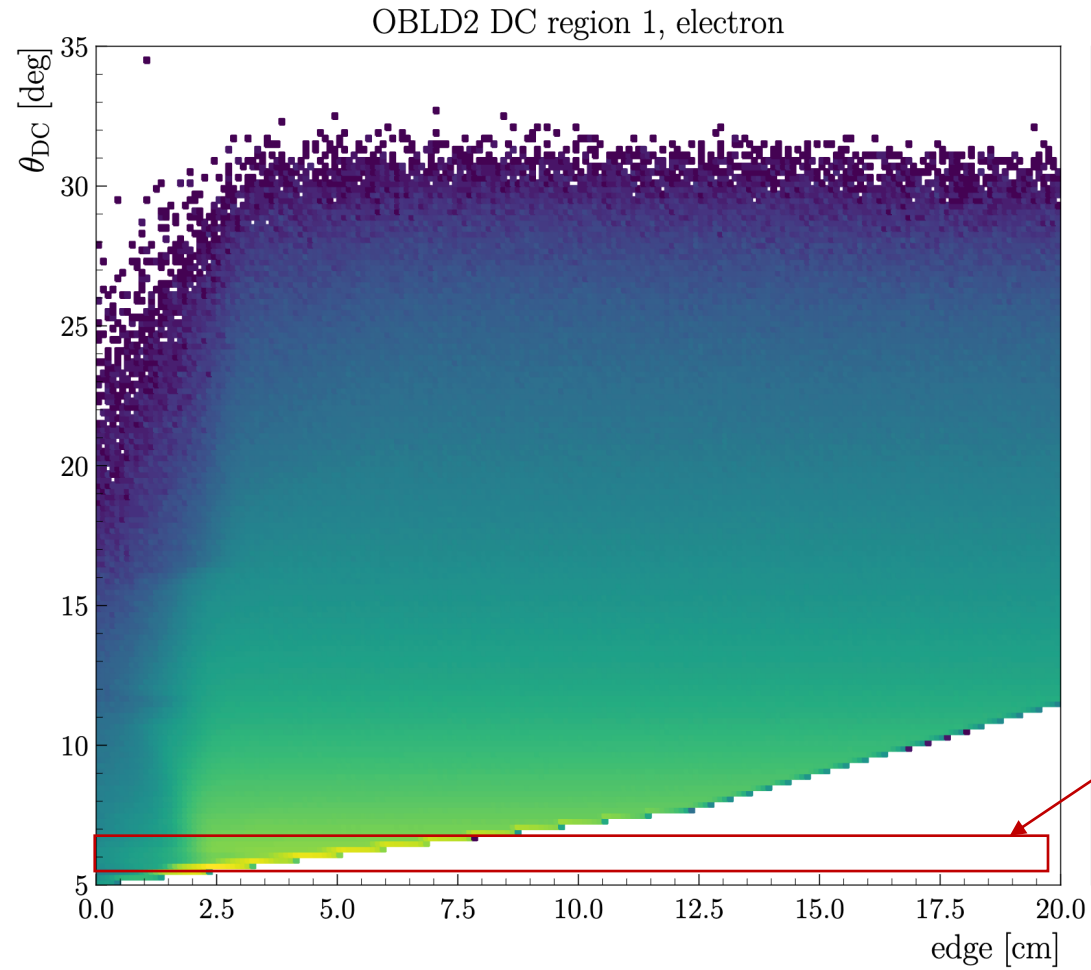
- REC::Traj:edge, distance between the trajectory point and the detector edge
- Just for the electron (pid=11)
- θ_{DC} : angle theta at the DC plan (x, y, z from the REC::Traj, sec from REC::Track bank.

```
auto Reader::compute_theta_local_DC(double x, double y, double z, int sec) -> double {
    // Rotate Z by PI/3*(sector-1)
    double angle = (sec - 1) * M_PI / 3;
    double s = std::sin(x: angle);
    double c = std::cos(x: angle);
    double x0 = x;
    double y0 = y;
    x = c * x0 - s * y0;
    y = s * x0 + c * y0;

    return std::atan2(y: std::sqrt(x: x * x + y * y), x: z) * 180.0 / M_PI;
}
```



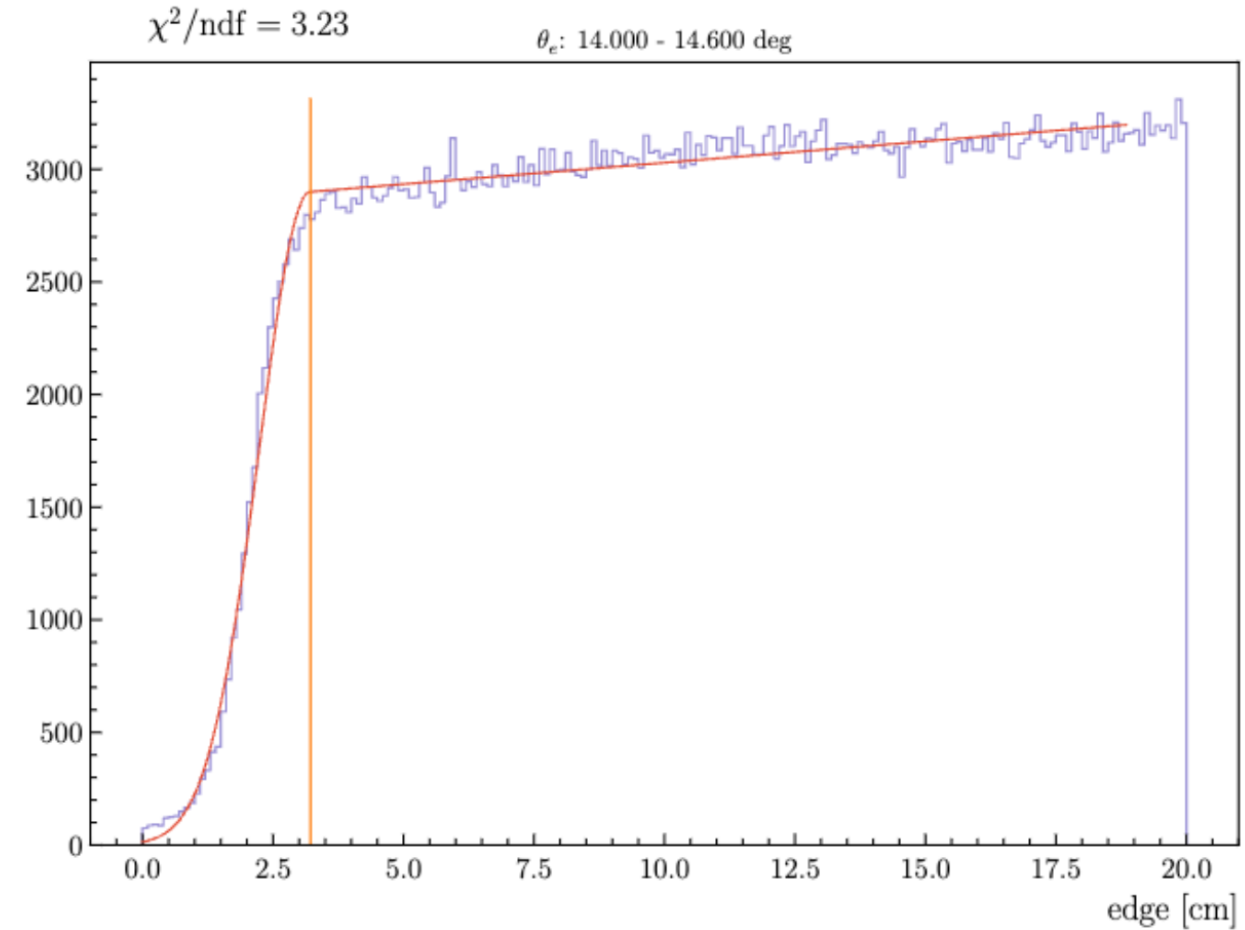




- Fitted with:
 - Gaussian from 0 to μ and a straight line after

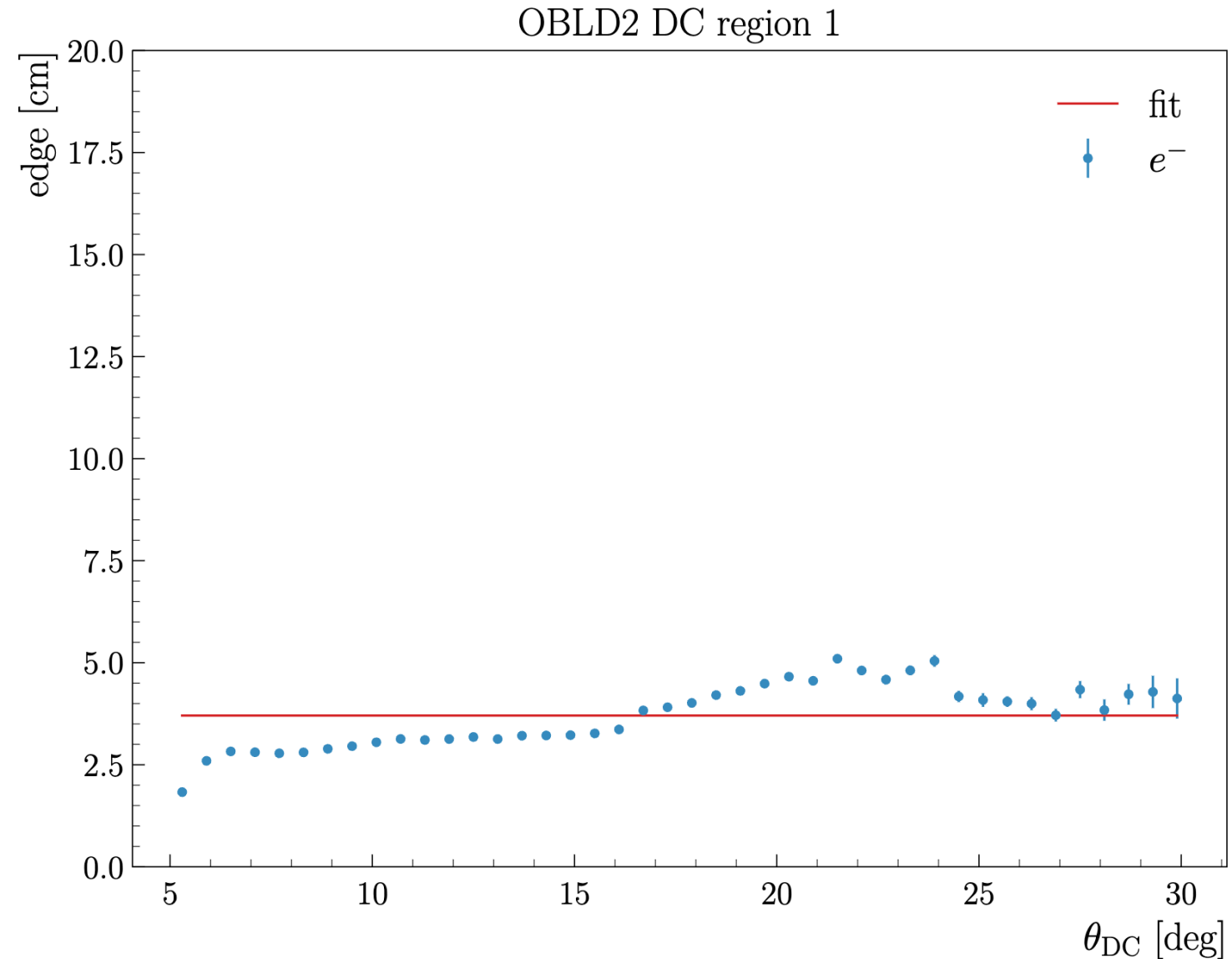
```
def fit_func(x: np.ndarray, pars: np.ndarray):  
    if np.abs(x[0]) < pars[1]:  
        return pars[0] * ROOT.TMath.Gaus(np.abs(x[0]), pars[1], pars[2])  
    else:  
        return pars[0] + (np.abs(x[0]) - pars[1]) * pars[3]
```

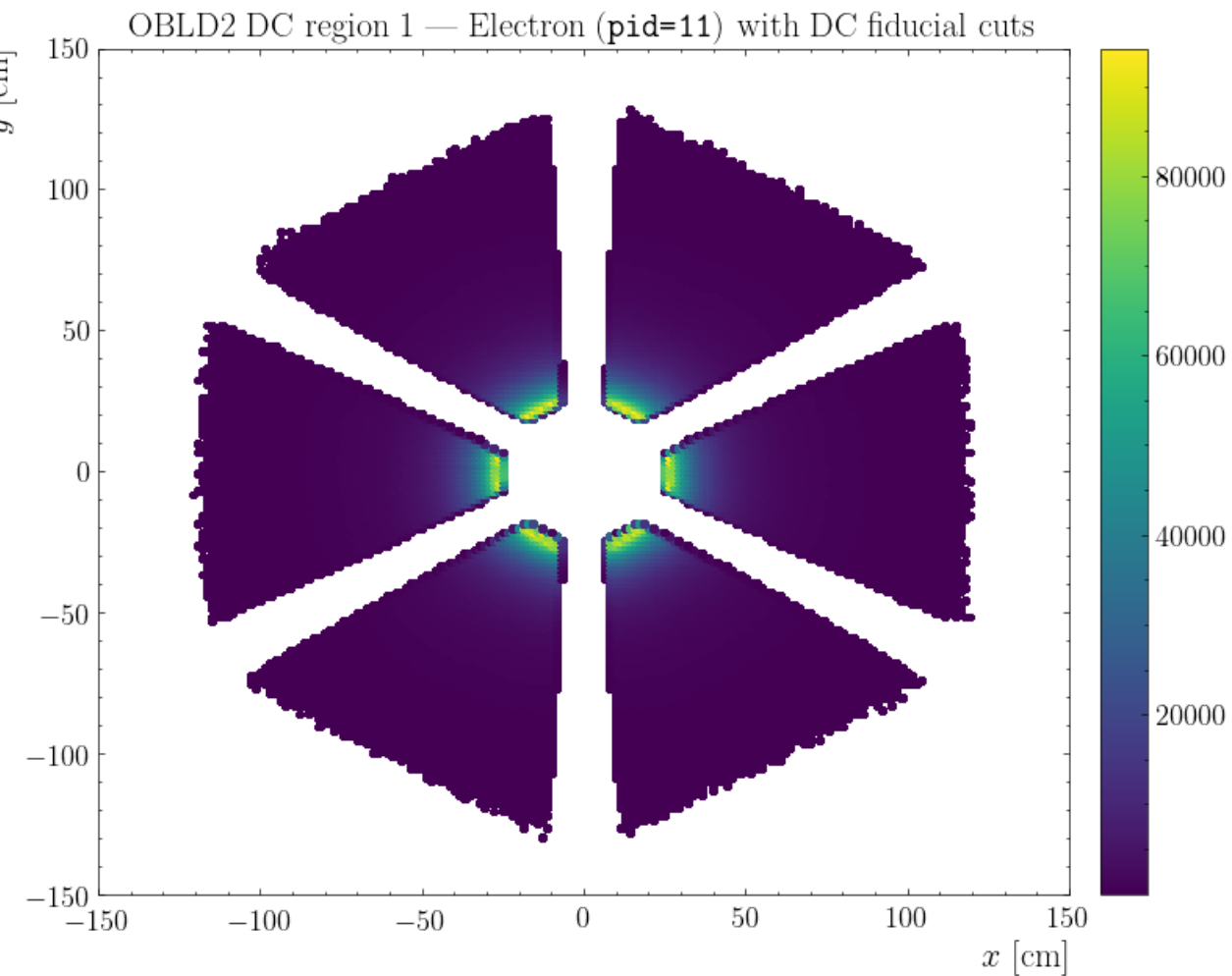
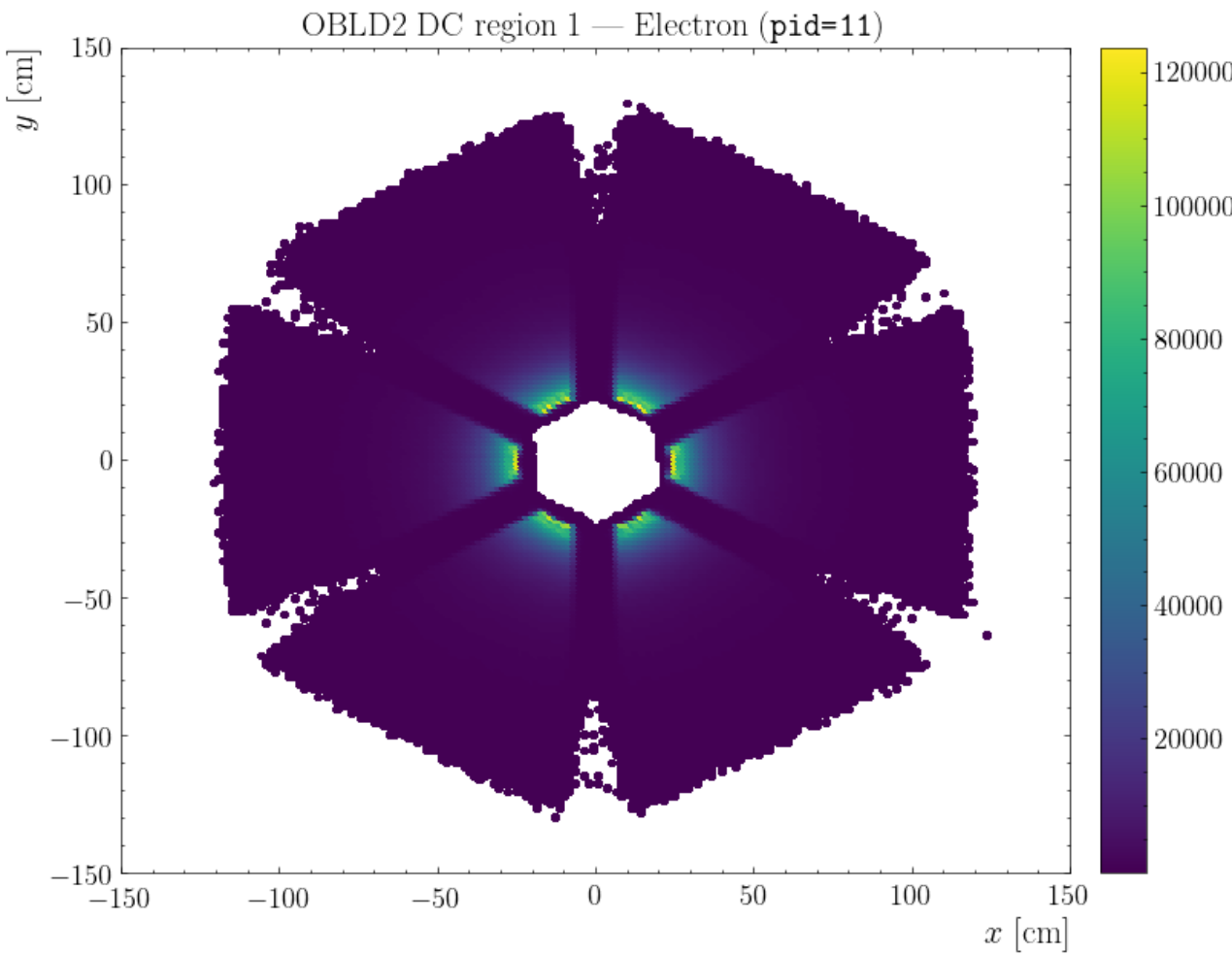
Extract the mean (μ)



RG-D: Fiducial cuts

- Plot the μ as a function of θ_{DC}
- Fit it with a constant:
 - 3.71 cm

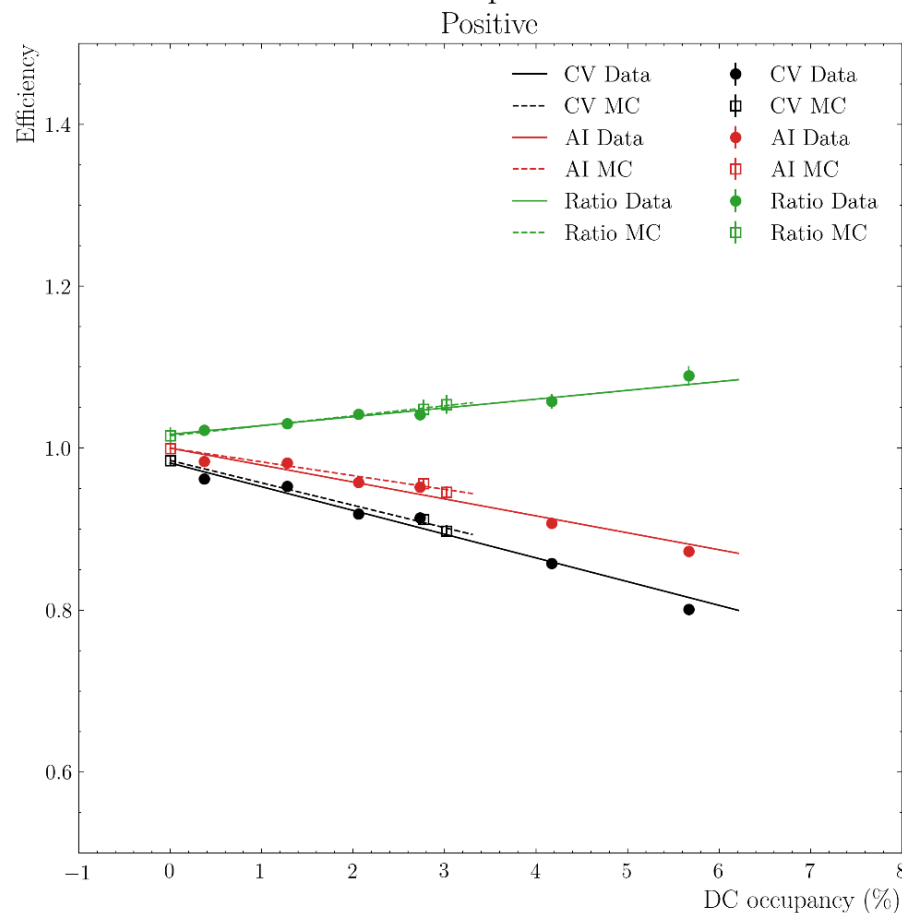




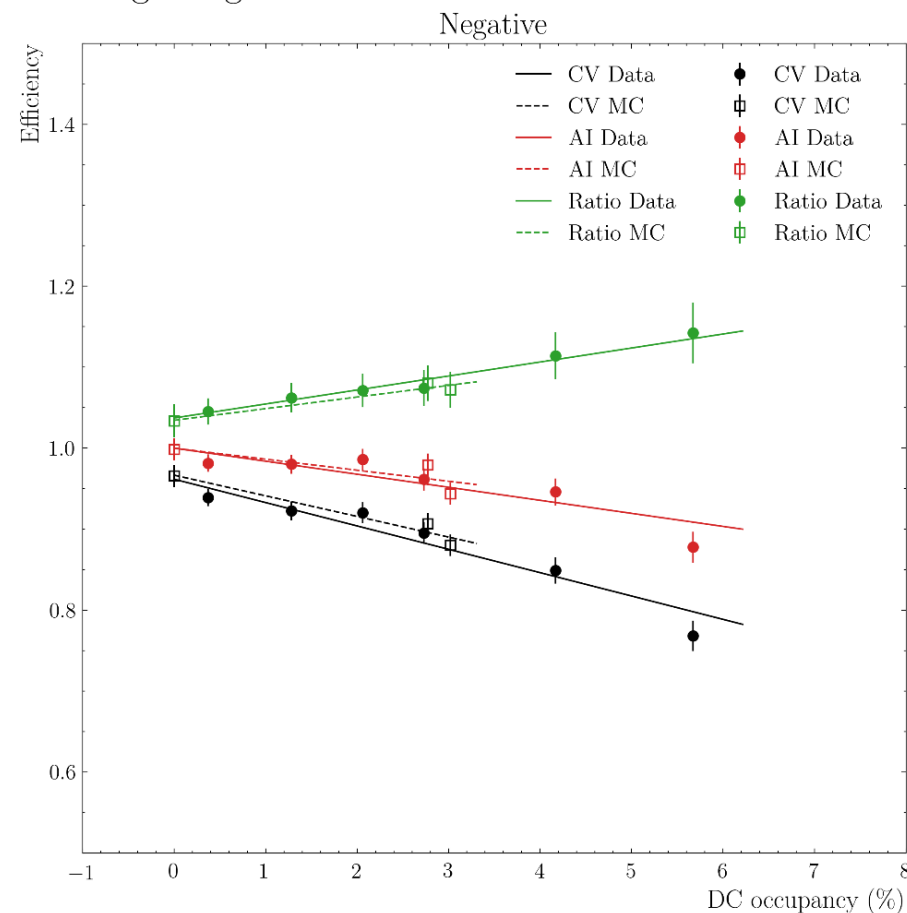
RG-D: Efficiency vs DC occupancy

- Using the decoded files of Pass0v8, extract the DC occupancy for a run (20 files):
 - Select the random events with the trigger (if `RUN::config:trigger & 0x0000008000000000 = true`)
 - DC occupancy per event = (number of hits in the `DC::tdc` bank) / (6 x 6 x 6 x 112) 6 sectors, 6 superlayers, 6 layers, 112 wires
 - Take the average for the 20 files

Comparison LD2 MC vs. Data with denoising using OBLD2 network

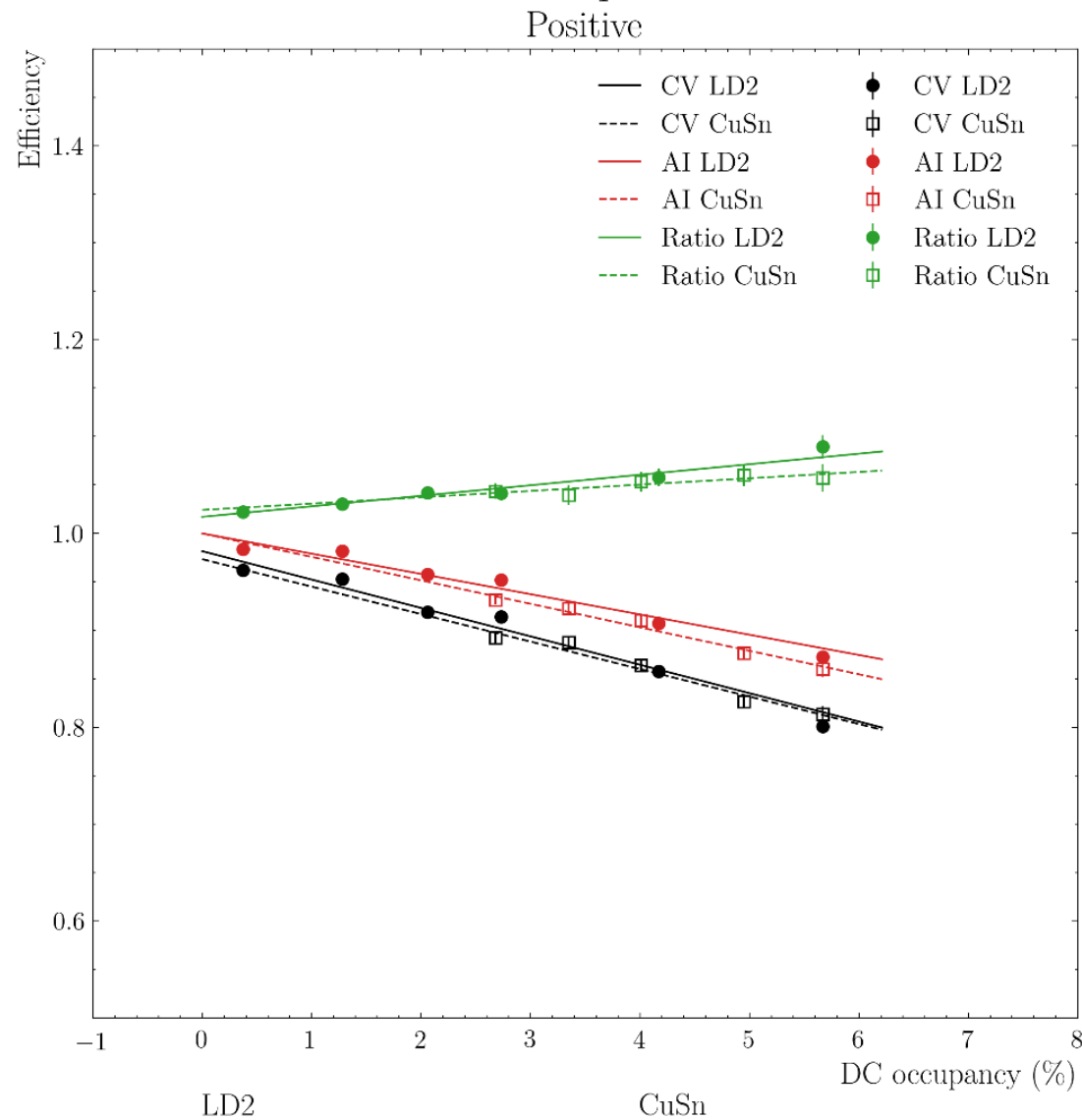


$1.00000 + -0.020909 \times (\pm 0.001082)$	$1.00000 + -0.016963 \times (\pm 0.003043)$
$0.98166 + -0.029299 \times (\pm 0.001136)$	$0.98481 + -0.027674 \times (\pm 0.003045)$

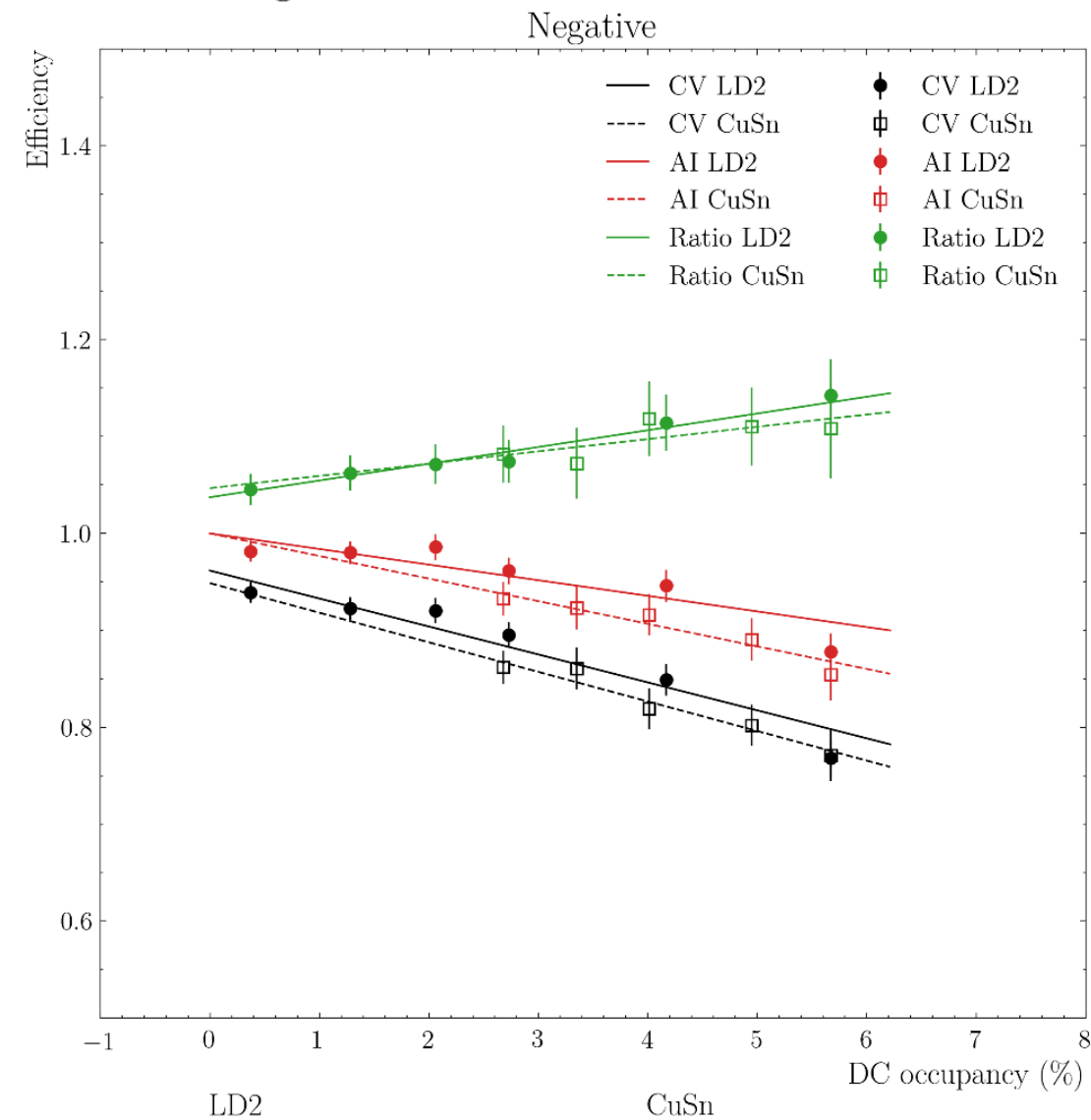


$1.00000 + -0.016103 \times (\pm 0.003451)$	$1.00000 + -0.013618 \times (\pm 0.005776)$
$0.96165 + -0.028847 \times (\pm 0.003411)$	$0.96665 + -0.025492 \times (\pm 0.005708)$

Comparison of IBCuSn and IBLD2 lumi runs using OBLD2NetDen



$$\begin{aligned} &1.00000 + -0.020909 \times (\pm 0.001082) & 1.00000 + -0.024227 \times (\pm 0.002598) \\ &0.98166 + -0.029299 \times (\pm 0.001136) & 0.97343 + -0.028359 \times (\pm 0.002719) \end{aligned}$$



$$\begin{aligned} &1.00000 + -0.016103 \times (\pm 0.003451) & 1.00000 + -0.023314 \times (\pm 0.009004) \\ &0.96165 + -0.028847 \times (\pm 0.003411) & 0.94867 + -0.030517 \times (\pm 0.008893) \end{aligned}$$

Task	expected	done	comment
Improve pid, fiducials cuts	04/21		Work on the DC fiducials, need to produce a PDF with all the plots and details, some issues with defining the DC limit. look good now
Produce physics plots for pass1 review	04/30		
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on real data (ALERT)	04/21	done	
Test the AI on ^4He	06/02		
Write the AI study on the wiki RG-D	05/12		Need to write the background merging part

To do list

13

Task	expected	done	comment
Efficiency vs occupancy plot for RG-D		done	
Train the AI with a momentum range close to that the elastic proton is	06/02		